PathMNIST Federated Learning: FedAvg vs FedProx Under Non-IID Label Noise

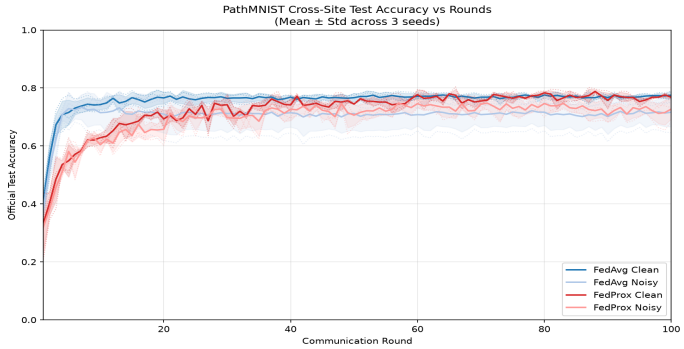
Authoritative Benchmark & Evaluation Report | Seeds: 42, 123, 2026 | Dirichlet α=0.3 | 40% Symmetric Noise on Client 0
GitHub: https://github.com/syedHasnain04/pathmnist-federated-learning
W&B; Dashboard: https://wandb.ai/iamsyedhasnain04-gprec/pathmnist-federated-learning

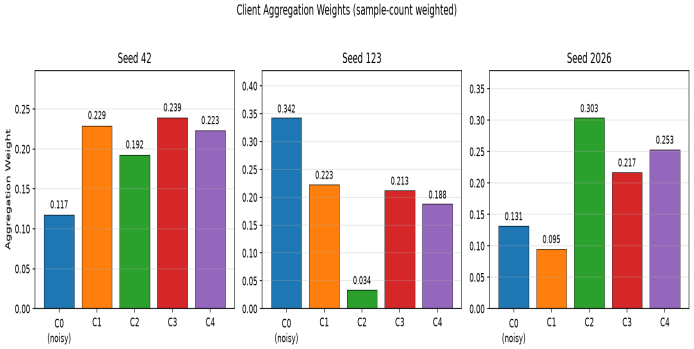
Experimental Design: 5-client Dirichlet (α=0.3) partitioning over official PathMNIST training data (89,996 images). Client 0 is subjected to 40% symmetric label noise on its local training set. 100 communication rounds, 5 local epochs (E=5), batch size 64, SGD optimizer (lr=0.01, momentum=0.9, fresh optimizer state per round). FedProx proximal term evaluated against detached start-of-round snapshot. Evaluation strictly isolates official in-distribution validation (10,004) from clinical cross-site test (7,180).

Table 1: Cross-Site Official Test & Validation Performance (Mean ± Std Across 3 Seeds)

Experimental Group	Test Acc (R100)	Val Acc (R100)	Test Acc (91-100)	Val Acc (91-100)	C0 Holdout (R100)
FEDAVG clean	0.7752 ± 0.0134	0.8485 ± 0.0022	0.7715 ± 0.0061	0.8488 ± 0.0017	0.8783 ± 0.0344
FEDAVG noisy	0.7164 ± 0.0501	0.7631 ± 0.0631	0.7149 ± 0.0459	0.7622 ± 0.0631	0.7514 ± 0.0641
FEDPROX clean	0.7684 ± 0.0052	0.8557 ± 0.0127	0.7673 ± 0.0025	0.8570 ± 0.0097	0.8849 ± 0.0232
FEDPROX noisy	0.7273 ± 0.0342	0.8278 ± 0.0221	0.7236 ± 0.0271	0.8248 ± 0.0234	0.8290 ± 0.0131

Key Visualizations: Convergence and Diagnostic Metrics





Update-Metric Observations & Robustness:
• *Noise Impact (FedAvg):* FedAvg clean test acc (rounds 91-100 mean): 0.7715; noisy: 0.7149. Noise degradation: +0.0566.
• *Noise Impact (FedProx):* FedProx clean: 0.7673; noisy: 0.7236. Noise degradation: +0.0437.
• *Algorithm Comparison (Noisy):* FedProx achieves higher mean test acc under noisy condition (difference: 0.0087).
• *Client 0 Aggregation Weight (p0):* seed 42: 0.117, seed 123: 0.342, seed 2026: 0.131. Higher p0 amplifies noise impact on the global model.
• *Update-Metric Analysis:* Client 0 (40% noisy) exhibits altered gradient dynamics. In non-IID partitioning (alpha=0.3), low cosine alignment occurs due to class imbalance. FedProx proximal term constrains per-client parameter drift from the round-start global snapshot. See plots 5-7 for per-client L2 norm and cosine similarity trajectories.

Limitations & Scientific Disclosures: (1) PathMNIST official test represents clinical cross-site evaluation; cross-site accuracy is lower than in-distribution validation. (2) μ was selected using in-distribution validation over rounds 91-100 on seed 42. (3) Three seeds provide preliminary variance estimation without formal statistical significance claims. (4) Offline JSON logging utilized as primary reproducible audit trail due to environment connectivity constraints.